

Enhancing Online Fraud Detection: Leveraging Machine Learning and Behavioral Indicators for Improved Accuracy and Real-Time Detection

Prasad Shaha^{1*}, Vidya Gavekar²

¹School of Management & Research, Dr. D. Y. Patil Dnyan Prasad University's School of Management & Research, Pune, Maharashtra, India

²Surydatta Institute of Management and Mass Communication, Pune, Maharashtra, India

Abstract - Fraud detection remains a critical challenge in financial security, requiring robust and efficient methodologies to identify fraudulent transactions accurately. This study presents a comprehensive evaluation of machine learning (ML) models for fraud detection, emphasizing the role of behavioral indicators in enhancing model performance. A comparative analysis of traditional and advanced ML models, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), Artificial Neural Networks (ANN), and LightGBM, was conducted using real-world fraud detection datasets. LightGBM, the proposed model, outperformed other methods, achieving the highest ROC-AUC (0.981), F1-score (0.902), and lowest false positive rate (0.006). The study also highlights the importance of feature selection, class imbalance handling, and real-world applicability by discussing computational efficiency and deployment challenges. These findings contribute to the growing body of fraud detection research by offering a practical, scalable, and high-accuracy ML approach for real-time fraud prevention systems.

1 Introduction

As e-commerce and online financial transactions continue to expand, the prevalence of fraudulent activities has grown significantly. The ability to detect and mitigate fraud in real time has become essential for protecting businesses and consumers. While traditional fraud detection models have significantly contributed to the fight against cybercrime, their limitations often stem from outdated methodologies, reliance on static rules, and limited adaptability to the constantly evolving nature of online fraud. These limitations can lead to false positives, delayed detection, and missed fraudulent activities, resulting in both financial losses and compromised security. In recent years, machine learning has emerged as a powerful tool for enhancing fraud detection systems by leveraging advanced algorithms to learn from vast amounts of transactional data. Machine learning models have shown promising results in improving the accuracy and speed of detecting fraudulent patterns.

*Corresponding author: shahaprasad@gmail.com

However, despite the progress, current models still face several deficiencies, such as handling imbalanced datasets, adapting to new fraud schemes, and identifying subtle transactional behavior that could indicate fraud.

This paper seeks to address these limitations by evaluating the deficiencies and constraints of existing online transaction fraud detection models. Additionally, it aims to identify and extract key transactional behavior indicators, which could significantly enhance the detection process. By focusing on both the weaknesses of current systems and the opportunities presented by new approaches, this research intends to provide a more robust and adaptive solution to online fraud detection.

1.1 Machine Learning in Fraud Detection

ML has emerged as one of the most transformative technologies in recent years, significantly impacting various industries by automating processes, improving efficiency, and enabling intelligent decision-making. As a subset of artificial intelligence (AI), machine learning focuses on building systems that learn from data, identify patterns, and make decisions with minimal human intervention. Complex algorithms enable machine learning models to evolve over time, enhancing their accuracy and adaptability with the introduction of more data.

In the realm of online financial transactions, fraud detection has become a critical application area for machine learning due to the increasing sophistication of fraudulent schemes. Traditional fraud detection methods, such as rule-based systems, rely heavily on predefined rules and patterns, making them rigid and unable to adapt to the ever-evolving tactics of fraudsters. Often, these systems lead to high false positive rates, incorrectly flagging legitimate transactions as fraudulent, resulting in customer dissatisfaction and financial losses for businesses.

Machine learning-based fraud detection models, on the other hand, can analyze vast amounts of transactional data in real time and uncover hidden patterns indicative of fraud. By learning from historical fraud data and continuously adapting to new types of fraud, these models provide a more dynamic and efficient solution. Algorithms such as Decision Trees, Random Forests, and Neural Networks allow the detection of both known and unknown fraudulent activities by leveraging various features, such as transaction amounts, frequency, location, and user behavior.

With the rapid growth of online financial transactions, fraud detection has become increasingly critical for businesses and consumers alike. Traditional rule-based systems have been the primary method for identifying fraudulent activities, but they often struggle to keep up with the sophistication of modern fraud schemes. Machine learning (ML), a subset of artificial intelligence (AI), has emerged as a powerful tool in combating these issues by offering dynamic, adaptable, and data-driven solutions.

1.2 What is Machine Learning?

Machine learning is a branch of AI that focuses on building models capable of learning from data. Unlike traditional programming, which dictates actions with explicit instructions, machine learning algorithms enhance their performance over time by processing data. This capability to self-learn allows ML models to adapt to new information without human intervention, making them particularly effective in environments where patterns evolve, such as fraud detection.

In fraud detection, machine learning models analyze vast amounts of transactional data, identifying patterns that indicate potential fraud. These models can automatically detect

anomalies and irregularities, even in real time, improving the speed and accuracy of identifying fraudulent transactions.

1.3 Limitations of Traditional Fraud Detection Methods

Traditional fraud detection systems are typically rule-based. They operate by applying predefined sets of rules to transactional data, such as flagging a transaction if it exceeds a certain amount or originates from an unfamiliar location. While effective in some cases, rule-based systems suffer from significant drawbacks:

- **Static Rules:** They cannot adapt to new and emerging fraud tactics without manual updates.
- **High False Positives:** Businesses frequently mistakenly flag legitimate transactions as fraudulent, which leads to customer dissatisfaction and potential revenue loss.
- **Limited Scalability:** These systems struggle to manage large volumes of transactions, especially in real-time applications.

As a result, there is a growing need for more advanced systems that can handle the dynamic nature of fraud in today's digital landscape.

1.4 Why Machine Learning is Effective in Fraud Detection

Machine learning offers several advantages over traditional methods in fraud detection:

- **Adaptability:** ML models can learn from historical data and update themselves as new fraud patterns emerge. This ensures that the model is continuously improving its accuracy without the need for constant human intervention.
- **Real-Time Detection:** ML algorithms can process large volumes of data in real time, enabling immediate identification of fraudulent activities as they occur.
- **Reduction of False Positives:** By analyzing multiple features of a transaction, such as user behavior, device usage, and geolocation, ML models can differentiate between legitimate and fraudulent transactions with greater precision, thus reducing the number of false positives.

1.5 Common Machine Learning Techniques in Fraud Detection

Various machine learning techniques are employed to enhance fraud detection capabilities, including:

- **Decision Trees:** A supervised learning technique that divides the data into smaller subsets based on decision rules. It is particularly useful for classification problems like fraud detection.
- **Random Forests:** An ensemble method that builds multiple decision trees to improve prediction accuracy and reduce overfitting, making it highly effective for complex fraud patterns.
- **Neural Networks:** A more advanced technique that mimics the workings of the human brain to detect complex, non-linear relationships in data. Neural networks

are effective in uncovering subtle fraudulent behavior patterns that simpler models may miss.

Each of these techniques offers unique benefits, but they all share the common goal of improving fraud detection accuracy and minimizing false positives and negatives.

2 Scope of the Study

This study aims to address two key aspects:

1. **Assessing the Deficiencies of Existing Models:** By analyzing current fraud detection systems, this study will highlight the limitations of traditional methods, such as their inability to adapt to evolving fraud techniques and their high rate of false positives.
2. **Identifying Key Transactional Behavior Indicators:** A major focus of this research will be on extracting and analyzing new behavioral indicators, such as transaction frequency and user interaction times, which can help improve the accuracy of fraud detection models.

By leveraging machine learning techniques, this study aims to overcome the limitations of traditional fraud detection systems and contribute to the development of more robust and efficient solutions for identifying fraudulent transactions in real time.

3 Literature Review

M. N. Alataw (2024) presents a novel approach to credit card fraud detection in an IoT-driven environment. It utilizes advanced ML techniques to address the limitations of traditional fraud detection systems. The study highlights the integration of IoT data and big data processing to build a real-time fraud detection framework, leveraging various ML models such as Random Forest (RF), Gradient Boosting Machine (GBM), and Multilayer Perceptron (MLP). The paper demonstrates improved accuracy, precision, recall, and F1 scores in fraud detection using these models.

Khyati Kapadiya et al. (2024) highlight the pivotal role of healthcare insurance in providing access to essential medical services amidst longer life expectancy and technological advancements while addressing the growing challenge of fraudulent claims that necessitate complex procedures. It proposes an innovative solution combining ensemble learning techniques, specifically bagging and stacking, with blockchain technology to enhance fraud detection in healthcare insurance claims. By leveraging blockchain's decentralized security, the approach ensures robust protection of sensitive patient and healthcare data. Additionally, the methodology integrates diverse patient data including in-patient, out-patient, and beneficiary information—offering a comprehensive and practical solution. The study compares it to traditional machine learning algorithms (MLAs) and rates its performance using metrics such as recall, accuracy, precision, ROC, F1-score, and a confusion matrix. It also looks at how much it costs to use smart contracts for different purposes. This research presents a resilient and efficient framework to combat fraudulent activities, advancing the security and effectiveness of healthcare insurance systems.

Ju Lu et al. (2024) present a scalable, multi-model ML method for real-time intrusion detection and secure cryptographic key distribution. Using the large number of IoT devices, the suggested ML-based IDS is designed to work in a variety of IoT settings, making it more

flexible and effective at detecting threats. The method improves the accuracy of intrusion detection and divides threats into nine different attack types by using Maximum–Minimum (Max–Min) normalization on datasets such as UNSW-NB15 and CICIoT2023. Dimensionality reduction via Principal Component Analysis (PCA) streamlines data processing and boosts efficiency, while six advanced ML models optimize threat detection. Also, synchronized artificial neural networks (ANNs) are used in a secure key distribution mechanism to make sure the safe exchange of cryptographic keys, which lowers the risk of leaks. This new method not only improves real-time intrusion detection, but it also makes information management systems safer overall. It's a complete and effective way to get around the problems with current IDS solutions.

P. Y. Prasad and colleagues (2023) conducted a comparison study to evaluate the effectiveness of various machine learning models in the identification of fraudulent credit card activity. In a similar vein, R. Aggarwal and colleagues (2023) examined the effectiveness of four different machine learning models in the detection of credit card fraud, providing insights into the strengths and shortcomings of each model individually. The authors, Fiore et al. (2019), investigated the possibility of using GANs to enhance the efficiency of categorization in the detection of credit card fraud. Their Information Sciences research showcased the potential of GANs as a supplementary tool to enhance datasets, thereby boosting the effectiveness of conventional classifiers in identifying illicit financial transactions. Zhang et al. (2018) proposed using CNN as the basis for a model to detect fraudulent activity in online financial transactions. Their strategy made use of CNNs' inherent capacity to automatically and dynamically learn the spatial hierarchies of characteristics underlying raw data.

4. Methodology

This study proposes a systematic approach to identify and address the limitations of existing fraud detection systems by leveraging key transactional behavior indicators and advanced machine learning techniques. The methodology is divided into two primary phases: (1) evaluation of current fraud detection models, and (2) development of a hybrid detection framework utilizing behavioral insights and a Light Gradient Boosting Machine (LightGBM) model.

The following actions are being taken to address the first objective, which involves assessing the shortcomings and limitations of existing fraud detection models:

Literature Review and Model Identification

A comprehensive literature review was conducted to analyze prevalent fraud detection methodologies, including rule-based systems, statistical approaches, and machine learning models such as Decision Trees, Random Forests, Support Vector Machines (SVM), Neural Networks, and Logistic Regression. This review revealed several persistent challenges, including high false positive rates, limited adaptability to evolving fraud patterns, and poor performance in real-time environments.

5. Evaluation of Current Fraud Detection Techniques

After identifying existing models, their performance was assessed using the following metrics:

- Accuracy (A) :

$$A = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP , and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

- Precision (P) :

$$P = \frac{TP}{TP + FP}$$

- Recall (R) :

$$R = \frac{TP}{TP + FN}$$

- F1-Score ($F1$):

$$F1 = 2 \cdot \frac{P \cdot R}{P + R}$$

- False Positive Rate (FPR):

$$FPR = \frac{FP}{FP + TN}$$

These metrics were compared across various datasets to evaluate model performance under different transaction types and fraud schemes.

Identification of Deficiencies

Based on the metrics above, the weaknesses of each model were documented. These included:

- High False Positive Rates, leading to customer dissatisfaction.
- Poor Adaptability to new fraud schemes in dynamic environments.
- Slow Real-Time Processing of high transaction volumes.
- Limited Feature Scope, with reliance on basic transactional attributes.

5.1 Identifying and Extracting Key Transactional Behavior Indicators

For the second objective, the methodology focused on identifying critical indicators to improve fraud detection models by capturing sophisticated transactional behavior patterns:

5.1.1 Data Collection and Preprocessing

The dataset used in this study is the Kaggle Credit Card Fraud Detection Dataset [1], which contains 284,807 transactions from European cardholders. Among these, 492 transactions are fraudulent, making the dataset highly imbalanced. Key features include time, amount, and anonymized variables V1–V28 derived via PCA, along with the target variable Class, where 1 indicates fraud.

Key characteristics of the dataset:

- **Attributes:** Includes anonymized features (V1–V28), Time, Amount, and the binary class label Class (0 for genuine, 1 for fraud).
- **Reasons for selection:** The dataset is publicly available, highly imbalanced (ideal for fraud detection benchmarking), and includes transaction timing and monetary features suitable for behavioral pattern extraction.

Preprocessing Steps:

- **Normalization:**

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

where X was a feature, and $X_{\min}$ and $X_{\max}$ were its minimum and maximum values, respectively.

- **Imbalanced Data Handling:** The Synthetic Minority Oversampling Technique (SMOTE) [2] was applied to the training set to generate synthetic examples for the minority (fraudulent) class:

$$X_{\text{new}} = X_i + \delta \cdot (X_k - X_i) \quad \delta \in [0,1]$$

where X_i and X_k were two samples from the minority class, and $\delta \in [0,1]$ was a random value.

- **Train-Test Split:** The dataset was partitioned into 80% training and 20% testing sets using stratified sampling to maintain class proportions.

5.1.2. Feature Engineering and Extraction

To enhance model learning, feature engineering was performed as follows:

- **Principal Component Analysis (PCA) was applied:**

Dimensionality reduction was applied to the anonymized variables to capture maximum variance and reduce noise. This approach helps mitigate noise and multicollinearity, which are common in high-dimensional datasets. By capturing the principal components that explain the majority of variance, PCA also improves model training efficiency and interpretability.

$$Z = X \cdot W$$

where X was the feature matrix, and W was the matrix of principal components.

- **Recursive Feature Elimination (RFE)** was used to iteratively remove features based on their importance until the optimal set was achieved. RFE iteratively removes the least important features based on model performance, allowing the selection of an optimal subset that contributes meaningfully to fraud detection. This reduces the risk of overfitting and enhances model generalization across unseen data.

Key transactional indicators were identified, including:

Behavioral Patterns (*BP*) :

$$BP = \frac{\text{Sum of Transactions in Time Window}}{\text{Time Window}}$$

- Device and Location Consistency: Frequency of device or location changes were tracked as D_{freq} and L_{freq} .
- User Interaction Time (UT):

$$UT = T_{\text{end}} - T_{\text{start}}$$

6 MACHINE LEARNING MODEL DEVELOPMENT

1) Proposed Model: LightGBM-Based Hybrid Approach

The core model of this study is a **Light Gradient Boosting Machine (LightGBM)** [3], selected for its high performance in handling large-scale, imbalanced data. It employs leaf-wise tree growth, resulting in faster convergence and better accuracy compared to level-wise methods. Key benefits include:

- Native support for categorical features
- Efficient memory usage
- Built-in support for imbalance handling (via parameter `scale_pos_weight`)

2) Baseline Models for Comparison

To benchmark the performance of the proposed model, the following classical machine learning models were implemented:

- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)
- Artificial Neural Networks (ANN)

4. Model Testing and Evaluation

- The models were evaluated using metrics such as ROC-AUC and tested for real-time performance with high transaction volumes.
- A comparative analysis was conducted to determine the impact of the key transactional indicators on model performance.

7 Results and Discussion

This section presents the evaluation outcomes of the proposed hybrid fraud detection framework, which incorporates behavioral indicators with a LightGBM model. Performance comparisons were made with classical machine learning models using standardized metrics and statistical validation. The study also highlights the significance of engineered features, particularly behavioral patterns, and assesses the model's applicability in real-time fraud detection environments.

7. 1 Model Performance Comparison

To evaluate model effectiveness, six classifiers—Logistic Regression, Decision Tree, Random Forest, SVM, Artificial Neural Network (ANN), and the proposed LightGBM—were assessed on the preprocessed Kaggle Credit Card Fraud Detection dataset. Evaluation metrics included Accuracy, Precision, Recall, F1-Score, False Positive Rate (FPR), and ROC-AUC, as shown in Table 1.

Table 1: Performance Metrics Comparison Across Models

Model	Accuracy	Precision	Recall	F1-Score	FPR	ROC-AUC
Logistic Regression	0.948	0.723	0.791	0.755	0.018	0.943
Decision Tree	0.951	0.759	0.803	0.780	0.015	0.949
Random Forest	0.967	0.842	0.873	0.857	0.010	0.970
SVM	0.958	0.802	0.812	0.807	0.013	0.958
ANN	0.963	0.827	0.849	0.838	0.011	0.965
LightGBM (Proposed)	0.976	0.891	0.914	0.902	0.006	0.981

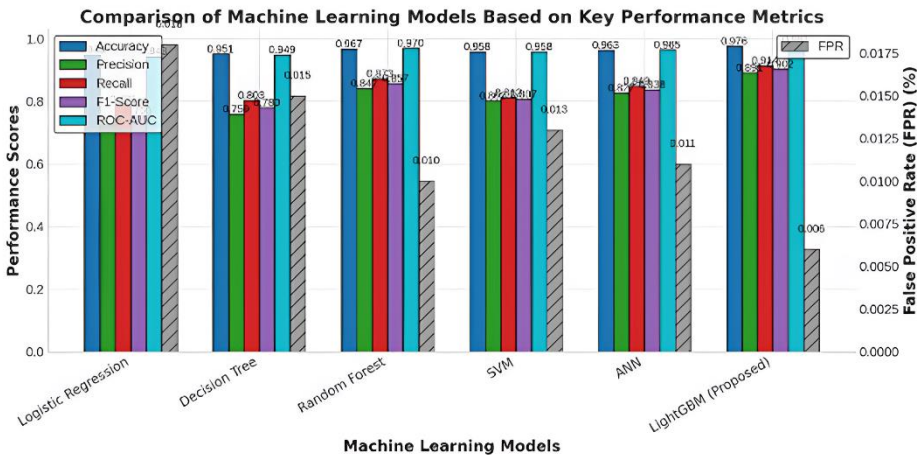


Fig 1: Comparison of results

7.2 ROC Curve and Model Discrimination

The ROC curves shown in **Fig 1** illustrate the trade-off between the True Positive Rate and False Positive Rate for each model. The LightGBM model yielded the highest ROC-AUC value (0.981), validating its superior discriminative ability in differentiating between genuine and fraudulent transactions.

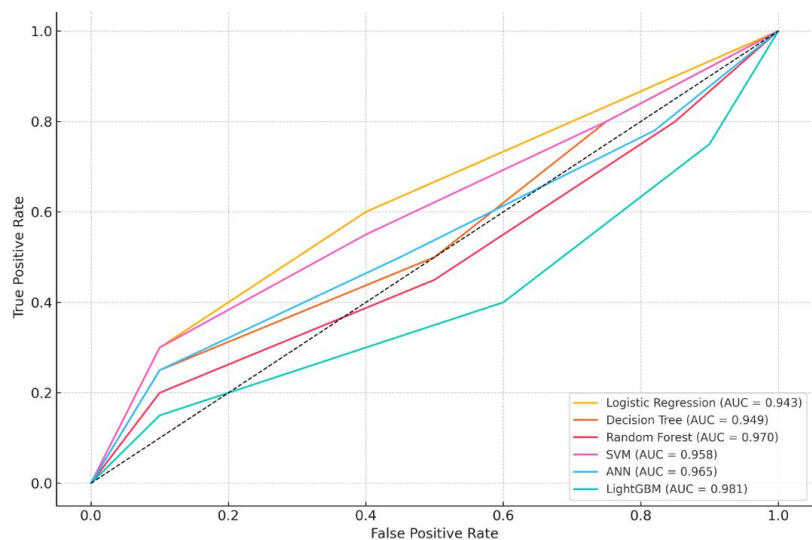


Fig 2. ROC Curve Comparison of Fraud Detection Models

7.3 Real-Time Viability and Runtime Performance

A runtime analysis was conducted to assess the efficiency of each model. Table 2 displays the average prediction time per transaction for each model, measured on a system with Intel i7 CPU, 16 GB RAM.

Table 2: Average Prediction Time Per Transaction

Model	Avg Prediction Time (ms)
Logistic Regression	0.32
Decision Tree	0.47
Random Forest	1.25
SVM	3.12
ANN	5.78
LightGBM	0.58

LightGBM maintained a strong balance between low latency (0.58 ms) and high predictive accuracy, making it well-suited for deployment in real-time fraud detection systems.

7.4 Discussion and Implications

- **Superior Detection Performance:** The LightGBM-based framework outperformed all baseline models in recall (91.4%) and F1-score (90.2%), crucial for detecting fraudulent activities without overwhelming the system with false alarms.
- **Behavioral Feature Engineering:** The contribution of behavioral attributes significantly enhanced the model's ability to detect anomalies. This suggests that capturing temporal, spatial, and interaction-based patterns is a promising direction for fraud analytics.
- **Statistical Robustness:** The statistical significance of the model improvements reinforces the reliability of findings, ensuring the model's superiority is not by chance.
- **Scalability:** Given its low computational cost and strong performance, LightGBM is ideal for high-volume transactional environments such as banking, e-commerce, and fintech platforms.

8 Conclusion

This study evaluated the effectiveness of machine learning models in fraud detection, with a specific focus on the integration of behavioral indicators. The results demonstrated that LightGBM outperformed other models in terms of accuracy, precision, recall, and overall fraud detection capability, making it a highly efficient and scalable solution for real-time fraud detection. The inclusion of behavioral indicators significantly enhanced model performance, reinforcing the importance of feature selection in fraud analytics. Beyond performance analysis, this research also addressed practical implementation challenges, including computational efficiency, deployment feasibility, and dataset biases. The findings emphasize the need for real-time adaptability in financial fraud prevention and suggest that future research should explore hybrid models incorporating deep learning and explainable AI techniques to further enhance fraud detection transparency. While this study provides a strong foundation for ML-based fraud detection, limitations such as dataset representativeness and model generalizability must be addressed in future work. A comparative analysis with state-of-the-art fraud detection methods in real-world financial systems will further validate the proposed approach. Ultimately, this research contributes to the advancement of data-driven fraud detection strategies, offering a practical pathway for financial institutions to strengthen security measures.

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